CS554 Project Report
Parallel OLS with Randomly Pivoted Cholesky for Sparse Approximation

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May 14, 2026

1 Introduction

In this project, we want to solve the following sparse approximation problem. Given $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, the sparse reconstruction problem aims to solve

$$\min_{x\in\mathbb{R}^n} \|Ax - b\|_2^2 \quad \text{subject to} \quad \|x\|_0 \le k.$$

Here $\|x\|_0$ denotes the number of nonzero entries of x. In this project, we are interested in a special case: the target vector is $b = Aw$, where w is a dense weight vector, and all columns in A are normalized.

We write $[n] = \{1, \dots, n\}$. For $S \subseteq [n]$, A_S denotes the submatrix of A with columns indexed by S, and A_j denotes the j-th column of A. The sparse approximation problem can be written as a subset-selection problem: choose $S \subseteq [n]$, with $|S| \le k$, and solve

$$\min_{x_S\in\mathbb{R}^{|S|}} \|A_S x_S - Aw\|_2^2.$$

Given a fixed subset S, the coefficient vector x_S is obtained from a standard linear least-squares problem.

2 Related work

The algorithm for the sparse approximation problem can be classified as two main families. The first one solves the continuous optimization problem by relaxing the ℓ_0 norm to ℓ_1 norm. The second one considers it as a subset selection problem, and build the pivot set S in an incremental way until $|S| = k$. Amoung all subset selection methods, there are three basic algorithms: Matching pursuit (MP), Orthogonal Matching Pursuit (OMP) and Orthogonal Least Squares (OLS). MP builds the support by repeatedly selecting the atom most correlated with the current residual. OMP uses a similar correlation-based selection rule, but recomputes the least-squares coefficients on the selected support after each step. OLS is a more direct greedy rule: it selects the atom that gives the largest one-step decrease in the refitted squared residual. As shown below, the OLS update can be written in Gram-matrix form and implemented as pivoted partial Cholesky with a sparse-approximation pivot score. This connection lets us borrow randomized and blocked pivoting ideas from Randomly Pivoted Cholesky [1].

3 Orthogonal Least Squares with Pivoted Partial Cholesky

Let $A \in \mathbb{R}^{m\times n}$ be the dictionary matrix, with all columns normalized. Let $G := A^\top A$ be the Gram matrix. For a fixed pivot set S, define the Nyström approximation

$$\widehat{G}(S) = G_{:,S}\,(G_{S,S})^\dagger\, G_{S,:},$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse. When the selected principal block $G_{S,S}$ is nonsingular, this coincides with the approximation produced by pivoted partial Cholesky using the same pivots. In standard pivoted partial Cholesky, the pivot is chosen as the column with the largest diagonal entry in the residual matrix. For the sparse reconstruction problem, we need a different pivot rule, which will be discussed in later sections.

For a fixed pivot set S, the optimal squared residual in approximating Aw is exactly the Gram-residual quadratic form induced by the Nyström approximation.

Proposition 1. *For any subset $S \subseteq [n]$, given $A \in \mathbb{R}^{m\times n}$ and $w \in \mathbb{R}^n$,*

$$\min_{x\in\mathbb{R}^{|S|}} \|Aw - A_S x\|_2^2 = w^\top (G - \widehat{G}(S))w.$$

3.1 Pivoting strategy

The proposition implies that the subset-selection objective can be equivalently written as

$$\min_{|S|\le k} \mathcal{E}(S), \qquad \mathcal{E}(S) := w^\top (G - \widehat{G}(S))w.$$

As a result, the problem we now need to focus on is how to build a good pivot set S. Let $S^{(0)} = \emptyset$, $\widehat{G}^{(0)} = 0$, and $R^{(0)} = G$. After k pivots, let $S^{(k)}$ denote the pivot set and define

$$\widehat{G}^{(k)} := \widehat{G}(S^{(k)}), \qquad R^{(k)} := G - \widehat{G}(S^{(k)}), \qquad \mathcal{E}_k := \mathcal{E}(S^{(k)}) = w^\top R^{(k)} w.$$

At iteration k, choose a pivot $s_{k+1} \in [n] \setminus S^{(k)}$ with $R^{(k)}_{s_{k+1},s_{k+1}} > 0$, and set $S^{(k+1)} = S^{(k)} \cup \{s_{k+1}\}$. Pivoted partial Cholesky updates the approximation and residual by the rank-one formulas

$$\widehat{G}^{(k+1)} = \widehat{G}^{(k)} + \frac{R^{(k)}_{:,s_{k+1}} R^{(k)}_{s_{k+1},:}}{R^{(k)}_{s_{k+1},s_{k+1}}}, \quad R^{(k+1)} = R^{(k)} - \frac{R^{(k)}_{:,s_{k+1}} R^{(k)}_{s_{k+1},:}}{R^{(k)}_{s_{k+1},s_{k+1}}}. \tag{3.1}$$

This formulation recovers the standard OLS pivot rule. Given the current support $S^{(k)}$, OLS chooses the next index by solving

$$s_{k+1} \in \operatorname*{arg\,max}_{i\in[n]\setminus S^{(k)}} \left(w^\top \frac{R^{(k)}_{:,i} R^{(k)}_{i,:}}{R^{(k)}_{i,i}} w \right) = \operatorname*{arg\,max}_{i\in[n]\setminus S^{(k)}} \frac{\left(R^{(k)}_{i,:} w\right)^2}{R^{(k)}_{i,i}}$$

This gives the standard greedy algorithm.
Inspired by [1], we can also consider a randomized alternative, where we define a pivot score g_i for each column

$$g_j^{(k)} := \begin{cases} w^\top \dfrac{R^{(k)}_{:,j} R^{(k)}_{j,:}}{R^{(k)}_{j,j}} w = \dfrac{\left(R^{(k)}_{j,:} w\right)^2}{R^{(k)}_{j,j}}, & j \notin S^{(k)} \text{ and } R^{(k)}_{j,j} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If $\sum_{\ell=1}^n g_\ell^{(k)} = 0$, then no further decrease is possible and the method stops. Otherwise, we sample the next pivot s_{k+1} with probability proportional to the pivot score, i.e.,

$$\mathbb{P}\{s_{k+1} = j \mid R^{(k)}\} = \frac{g_j^{(k)}}{\sum_{\ell=1}^n g_\ell^{(k)}}.$$

3.2 Convergence Rate

In the following paragraph, we give the theoretical convergence rate of the orthogonal least squares with pivoted partial Cholesky. All the proofs are deferred to Appendix A.

Theorem 1 (Greedy pivoting)**.** *Assume the columns of A are normalized and $w \neq 0$, and let $S^{(k)}$ be generated by the greedy OLS pivot rule*

$$s_{k+1} \in \operatorname*{arg\,max}_{i \in [n] \setminus S^{(k)}} g_i^{(k)}.$$

Then, for every $k \geq 0$,

$$\mathcal{E}_k \leq \frac{\mathcal{E}_0}{1 + k\mathcal{E}_0/\|w\|_1^2} \leq \frac{\|w\|_1^2}{k+1}.$$

Theorem 2 (Randomized pivoting)**.** *Assume the columns of A are normalized and $w \neq 0$, and let $S^{(k)}$ be generated by the randomized pivot rule with probabilities proportional to $g_j^{(k)}$, using the stopping convention above if all scores vanish. Then, for every $k \geq 0$,*

$$\mathbb{E}[\mathcal{E}_k] \leq \frac{\mathcal{E}_0}{1 + k\mathcal{E}_0/(n\|w\|_2^2)} \leq \frac{n\|w\|_2^2}{k+1}.$$

Since $\|w\|_1^2 \leq n\|w\|_2^2$, the greedy bound is tighter than the randomized version.

In summary, both pivot rules have an $O(1/k)$ squared-residual bound under column normalization. The greedy rule gives deterministic bounds and is at least as strong in these estimates: $\|w\|_1^2 \leq n\|w\|_2^2$ for the unconditional rates. The randomized rule gives bounds in expectation, with the extra factor appearing in the convergence constants.

3.3 Numerical Tests

We plot the spectrum of dictionary matrix A and its corresponding convergence result between different strategies in Figure 1.

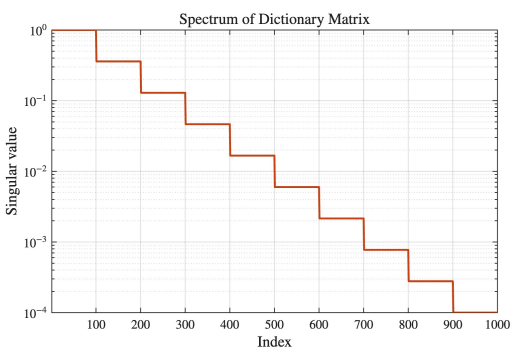
Figure 1a shows that the dictionary matrix $A \in \mathbb{R}^{5000 \times 1000}$ has a blockwise-flat, geometrically decaying singular spectrum. The singular values are grouped into several plateaus, with sharp spectral gaps between consecutive groups. The largest singular value is about 1, while the smallest is about 10^{-4}, so the matrix is ill-conditioned with condition number approximately 10^4.

In figure 1b, for each pivoting strategy, we run 10 independent trials. In each trial, we record the relative residual history

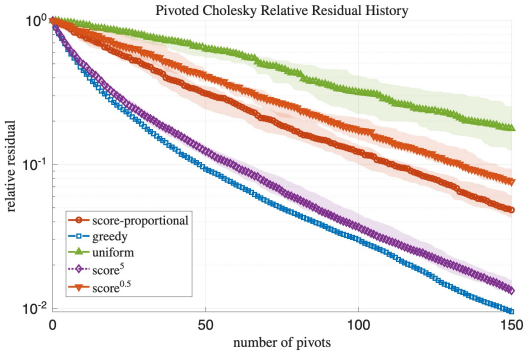
$$\frac{\mathcal{E}_k}{\mathcal{E}_0}, \qquad k = 0, 1, \ldots, K.$$

For every fixed pivot k, we compute the median relative residual over the 10 trials and plot it as the main curve. The shaded region is the empirical envelope over the trials: its lower boundary is the pointwise minimum residual and its upper boundary is the pointwise maximum residual. Thus, the shaded region represents the variability of the randomized pivoting rule, not a theoretical confidence interval or convergence bound.

Comparing several pivot selection rules for OLS pivoted Cholesky, all methods decrease the relative residual, but the convergence speeds differ significantly. The greedy rule gives the fastest convergence because it maximizes the one-step residual decrease at every iteration. The g_j^5 sampling rule closely tracks greedy behavior because the power transformation concentrates probability mainly on the largest OLS scores. The score-proportional ruleg_j is slower but corresponds to the theoretically justified randomized OLS pivot rule. The $g_j^{0.5}$ rule flattens the sampling distribution and therefore converges more slowly. Uniform sampling performs worst because it ignores the residual-dependent OLS score. Since the curves are nearly linear on the semilog plot, the empirical convergence is approximately geometric over this range of pivot ranks.



(a) Spectrum of dictionary matrix A



(b) Median relative residual $\mathcal{E}_k/\mathcal{E}_0$ over 10 trials for different pivoting strategies.

Figure 1: Convergence comparison on a dictionary matrix with clustered geometric spectrum.

4 Parallelize OLS with Pivoted Partial Cholesky

4.1 Rejection Sampling

In [1], a naive block RPCholesky algorithm is proposed, where a block of pivots is generated, and then the Cholesky factor is updated once. However, this naive algorithm can fail as it generates pivots from a stale distribution that should have been updated as each pivot is generated. This problem is fixed in a later work [2] by rejection sampling. We can apply the same idea to our OLS algorithm.

Given any residual matrix R, we define

$$q := Rw, \qquad d := \mathrm{diag}(R).$$

For $j \notin S$, the OLS score is

$$g_j = \begin{cases} q_j^2/d_j, & d_j > 0, \\ 0, & d_j = 0, \end{cases}$$

The sequential randomized OLS rule samples the next pivot with probability proportional to g_j. The purpose of the rejection step below is to recover this sequential score-proportional law while proposing a block of candidates at once.

Let $\Omega := \{j \in [n] \setminus S : d_j > 0\}$ be the currently active set. At the beginning of a block, define a proposal score

$$v_j = \begin{cases} g_j + c, & j \in \Omega, \\ 0, & j \notin \Omega, \end{cases}$$

where $c > 0$ is a small constant. This additive term gives the proposal distribution positive support on every active column. This is important because, unlike the residual diagonal in RPCholesky, an OLS score that is zero at the beginning of a block can become positive after another pivot in the same block has been accepted. If one wants a proposal distribution with uniform mixture weight $\alpha \in (0, 1)$, one may take

$$c = \frac{\alpha}{1-\alpha} \frac{\sum_{j\in\Omega} g_j}{|\Omega|},$$

which gives

$$\frac{v_j}{\sum_{\ell\in\Omega} v_\ell} = (1-\alpha)\frac{g_j}{\sum_{\ell\in\Omega} g_\ell} + \alpha\frac{1}{|\Omega|}.$$

If $\sum_{j\in\Omega} g_j = 0$, then no further OLS decrease is possible and the pivoting procedure stops.

Once candidates are proposed, we need to decide whether to accept them based on the updated true score. This can be done within the proposed block. Assume that an ordered set $T \subseteq \Omega$ of candidates has

already been accepted. The residual that would be obtained by sequentially adding the pivots in T is

$$R^{\langle T\rangle} = R - R_{:,T}(R_{T,T})^{\dagger}R_{T,:}.$$

Hence the true score of a remaining index $j \in \Omega \setminus T$ is

$$g_j^{\langle T\rangle} = \frac{\left(q_j - R_{j,T}(R_{T,T})^{\dagger}q_T\right)^2}{d_j - R_{j,T}(R_{T,T})^{\dagger}R_{T,j}},$$

with the convention that the score is zero if the denominator is nonpositive. Thus, the score update only needs the old vectors q, d and the residual columns indexed by the accepted pivots. As shown by the formula above, computing the updated score requires only at most a submatrix $n \times b$ of R.

4.2 1D Block Parallel OLS with Centralized Exact Rejection Sampling

In this section, we describe a parallel OLS algorithm with rejection sampling on a 1D processor grid.

The main processor stores full copies of q and d. Processor p owns an index block $\mathcal{I}_p \subset [n]$, with $\{\mathcal{I}_p\}$ forming a partition of $[n]$, and stores the corresponding rows of F and the corresponding local part of the score data.

Candidate Proposal At each round, we start by proposing a block of b candidates using the inactive-safe proposal weight v_j defined in the previous section. Processor p forms its local proposal mass $V_p = \sum_{j\in\mathcal{I}_p} v_j$, then one all-gather is performed to compute the total mass $V = \sum_p V_p$ as the normalization constant for the proposal probability. Once each processor have V, then propose candidates locally. The resulting candidates are sent to the main processor (replacements are removed). Let the candidate set be $C = (c_1, \ldots, c_b)$. Each processor should also compute and send the proper part of the current residual matrix

$$R_{\mathcal{I}_p,C} = G_{\mathcal{I}_p,C} - F_{\mathcal{I}_p,:}F_{C,:}^{\top}$$

Local Rejection Now the main processor has q, d and $R_{:,C}$ and needs to decide whether each candidate should be accepted or rejected. Let T be the accepted pivots within this block, and $\widetilde{q}$ and $\widetilde{d}$ be the updated q and d according to accepted pivots in T. While constructing T, the main processor also maintains the temporary Cholesky panel Y, such that $R^{(T)} = R - YY^{\top}$. We start with $T \leftarrow \emptyset$, $\widetilde{q} \leftarrow q$, $\widetilde{d} \leftarrow d$ and $Y \leftarrow [\,]$.

We iterate over each candidate in C. To maximize the acceptance rate, we need to compute the tightest coefficient M for the envelope first. Because the main processor has the full vectors $\widetilde{q}$ and $\widetilde{d}$, this can be done by a local scan:

$$M_T = \max_{j\notin S\cup T} \frac{\widetilde{g}_j}{v_j}, \qquad \widetilde{g}_j = \frac{\widetilde{q}_j^2}{\widetilde{d}_j}.$$

For a proposal $s \in C$, its true score is $\widetilde{g}_s = \widetilde{q}_s^2/\widetilde{d}_s$, and the proposal is accepted with probability

$$p_s = \frac{\widetilde{g}_s}{M_T v_s}$$

If s is rejected, no update is required. If it is accepted, we need to update the Cholesky panel Y, as well as $\widetilde{q}$ and $\widetilde{d}$. The update process is as followed:

1. Compute the new column y in the Cholesky panel and update $Y \leftarrow [Y\, y]$:

$$y = \frac{R_{:,s} - YY_{s,:}^{\top}}{\sqrt{\widetilde{d}_s}}$$

2. Update $\widetilde{q}$ and $\widetilde{d}$ ($\circ$ is element-wise product):

$$\widetilde{q} \leftarrow \widetilde{q} - \frac{y\widetilde{q}_s}{\sqrt{\widetilde{d}_s}}, \quad \widetilde{d} \leftarrow \widetilde{d} - y \circ y$$

Distributed Panel Update Once the main processor has gone through all candidates, it should have up-to-date $\widetilde{q}$, $\widetilde{d}$ and a Cholesky panel Y. The corresponding part of the panel Y_p is sent to processor p, and each processor can update the local Cholesky factor $F_{\mathcal{I}_p,:} \leftarrow [F_{\mathcal{I}_p,:} \quad Y_p]$.

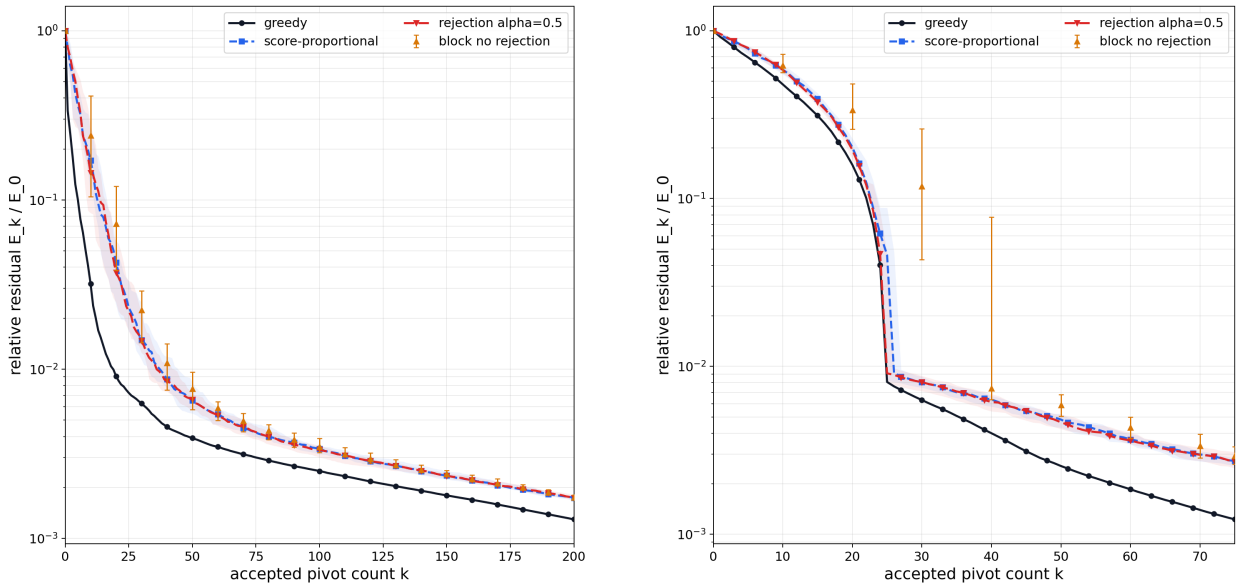
4.3 Numerical Tests

4.3.1 Convergence result

We first plot the convergence result in Figure 2. We test two different dictionary matrices A.

In the first test case, A is 400×400. For i from 1 to 50, the singular values A is set to be $\sigma_i = \exp(-i/10)$. For i from 50 to 400, $\sigma_i = \sigma_{50} \approx 7 \times 10^{-3}$. We can see from Figure 2a that our rejection sampling method have almost identical convergence rate compared to the sequential version, which implies that we have successfully recovered the distribution by rejection sampling. Greedy method still has a better convergence rate, as expected. In this testcase, there is no significant difference between the naive block algorithm without rejection and the rejection sampling method.

In the second test case, we construct the dictionary matrix A as follows. We first generate 25 random vectors $c \in \mathbb{R}^{2000}$. Then we add random Gaussian noise ϵ to each c, with $\|\epsilon\|_2 = 0.01\|c\|_2$ to get 50 different sample points for each center vector. We use these sample points as columns in A. In other words, we create 25 clusters, each consisting of near-duplicate samples. As is shown in Figure 2b, the greedy method, the randomized sequential method, and the randomized rejection sampling method give similar convergence rates before $k = 25$. This indicates that all these methods can successfully pick one sample point from each cluster without replacement. However, the naive block algorithm without rejection has a significantly worse convergence rate here. It suffers from the stale distribution score and wastes the pivots selecting near-duplicate points within the same cluster.



(a) The singular values of the dictionary matrix have exponential decay followed by a flat tail.

(b) The dictionary matrix contains several clusters, each consisting of near-duplicate samples.

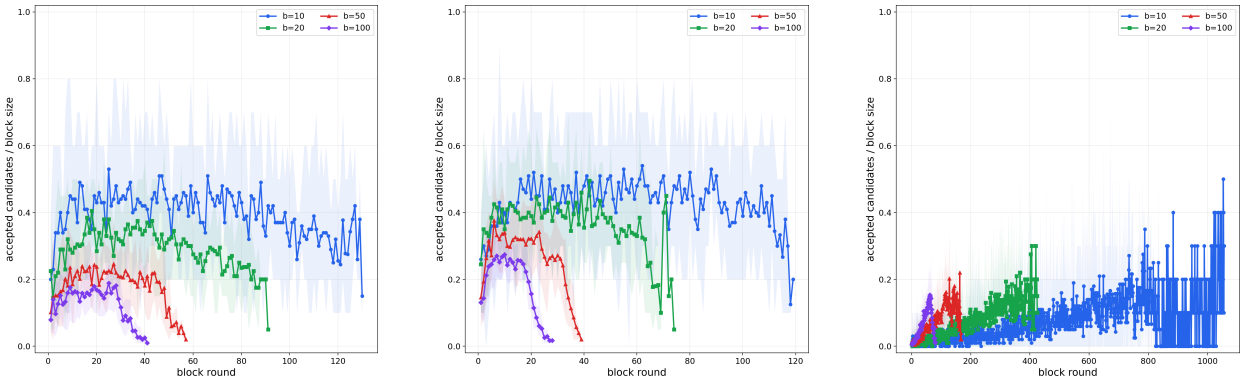
Figure 2: Convergence result of relative error $\mathcal{E}_k/\mathcal{E}_0$ against number of pivots.

4.3.2 Practical Tradeoffs

We also want to see how the block size and proposal distribution affect the acceptance rate. A higher acceptance rate means fewer candidates are wasted, and our algorithm works more efficiently. Throughout the section, we use the following setting: Set $A \in \mathbb{R}^{500\times500}$, the entries of the dictionary matrix are drawn

from a standard Gaussian distribution, $A_{ij} \sim N(0, 1/m)$. Then every column is normalized: $a_j \leftarrow a_j/\|a_j\|_2$. The weight vector is $w_j = j^{-1}$ for $j = 1, \ldots, n$

In Figure 3, we plot the acceptance rate against the index of rounds we propose candidates.



(a) $\alpha = 0.1$ (b) $\alpha = 0.5$ (c) $\alpha = 1$ (Uniform)

Figure 3: Acceptance rate of proposed candidates.

Proposal Distribution As mentioned in the previous section, zero-score columns can have positive scores after updating, so we need to mix our proposal probability with a uniform distribution. We use proposal score $v_j = g_j + c$, where g_j is the current pivot score, and c is a constant. We set

$$c = \frac{\alpha}{1-\alpha} \frac{\sum_{j\in\Omega} g_j}{|\Omega|},$$

This gives a mix between uniform and score-proportional distribution:

$$p_j = \frac{v_j}{\sum_{\ell\in\Omega} v_\ell} = (1-\alpha)\frac{g_j}{\sum_{\ell\in\Omega} g_\ell} + \alpha\frac{1}{|\Omega|}.$$

We set α to be $0.1, 0.5$, and 1 respectively.

By comparing the lines of the same color in all three subplots in Figure 3, we can see how different α affects the acceptance rate. We can see that $\alpha = 0.5$ gives the highest acceptance rate. The uniform sampling is bad because it does not take advantage of the information from the stale distribution effectively. When α is too small, the algorithm can also suffer from abrupt changes in the pivot score.

Block size We can also see how the block size affects the acceptance rate by focusing on each sub-figure and comparing lines in different colors. Generally, a larger block size means more parallelism. However, a larger block size can also decrease the acceptance rate because candidates are generated from a stale distribution that deviates further from the real distribution. According to the result shown in Figure 3, the results match our prediction. If we compute the number of accepted pivots, a larger block size does yield more accepted pivots, which is good news.

5 Conclustion

In this project, we apply the randomly pivoted Cholesky algorithm to the Orthogonal Least Squares method for the sparse approximation problem. We derived the theoretical convergence rate for randomized OLS, showing that it has the same asymptotic convergence rate as the classic greedy method. Though the greedy method is stronger in theory, the randomized method is more parallelizable. We developed a parallel version of randomized OLS on a 1D processor grid that uses block proposals for parallelism, but still recovers the correct distribution by rejection sampling. We test the algorithm with different test cases with various parameters.

References

[1] Yifan Chen, Ethan N. Epperly, Joel A. Tropp, and Robert J. Webber. Randomly pivoted cholesky: Practical approximation of a kernel matrix with few entry evaluations. *Communications on Pure and Applied Mathematics*, 78(5):995–1041, 2025.

[2] Ethan N. Epperly, Joel A. Tropp, and Robert J. Webber. Embrace rejection: Kernel matrix approximation by accelerated randomly pivoted cholesky, 2025.

A Proofs

Proof of Proposition 1. Let Π_S denote the orthogonal projector onto $\mathrm{col}(A_S)$. Then

$$\min_{x\in\mathbb{R}^{|S|}} \|Aw - A_S x\|_2^2 = \|(I-\Pi_S)Aw\|_2^2.$$

Since Π_S is an orthogonal projector, it is symmetric and idempotent, hence

$$\|(I-\Pi_S)Aw\|_2^2 = w^\top A^\top (I-\Pi_S)Aw.$$

Therefore,

$$\min_{x\in\mathbb{R}^{|S|}} \|Aw - A_S x\|_2^2 = w^\top A^\top (I-\Pi_S)Aw.$$

It remains to identify $A^\top \Pi_S A$. By the formula for the orthogonal projector onto $\mathrm{col}(A_S)$,

$$\Pi_S = A_S A_S^\dagger = A_S (A_S^\top A_S)^\dagger A_S^\top.$$

Hence

$$A^\top \Pi_S A = A^\top A_S (A_S^\top A_S)^\dagger A_S^\top A.$$

Using $G = A^\top A$, we obtain

$$A^\top A_S = G_{:,S}, \qquad A_S^\top A_S = G_{S,S}, \qquad A_S^\top A = G_{S,:},$$

and thus

$$A^\top \Pi_S A = G_{:,S}\,(G_{S,S})^\dagger\, G_{S,:} = \widehat{G}(S).$$

Substituting this identity into the previous expression gives

$$A^\top (I-\Pi_S)A = G - \widehat{G}(S),$$

so that

$$\min_{x\in\mathbb{R}^{|S|}} \|Aw - A_S x\|_2^2 = w^\top (G - \widehat{G}(S))w.$$

This proves the result. □

Proof of Theorem 1. Let $\Pi_{S^{(k)}}$ denote the orthogonal projector onto $\mathrm{col}(A_{S^{(k)}})$, and define

$$r_k = (I-\Pi_{S^{(k)}})Aw.$$

By Proposition 1, $\mathcal{E}_k = \|r_k\|_2^2$. Moreover,

$$R^{(k)} = A^\top (I-\Pi_{S^{(k)}})A,$$

so

$$R^{(k)}_{j,:} w = \langle (I-\Pi_{S^{(k)}})A_j, r_k\rangle, \qquad R^{(k)}_{j,j} = \|(I-\Pi_{S^{(k)}})A_j\|_2^2.$$

Thus $g_j^{(k)}$ is exactly the decrease in $\mathcal{E}_k$ if j is selected as the next pivot. Under the greedy rule,

$$\mathcal{E}_{k+1} = \mathcal{E}_k - \max_{j\in[n]} g_j^{(k)}.$$

It remains to lower-bound the maximum score. Since $r_k \perp \mathrm{col}(A_{S^{(k)}})$,

$$\mathcal{E}_k = \langle r_k, Aw\rangle = \sum_{j=1}^n w_j \langle r_k, A_j\rangle = \sum_{j=1}^n w_j \langle r_k, (I-\Pi_{S^{(k)}})A_j\rangle.$$

Using the definition of $g_j^{(k)}$, the convention $g_j^{(k)} = 0$ when $R^{(k)}_{j,j} = 0$, and the column normalization $\|A_j\|_2 = 1$, we have

$$|\langle r_k, (I-\Pi_{S^{(k)}})A_j\rangle| \le \sqrt{g_j^{(k)}}.$$

Therefore,

$$\mathcal{E}_k \le \sum_{j=1}^{n} |w_j| \sqrt{g_j^{(k)}} \le \|w\|_1 \sqrt{\max_{j\in[n]} g_j^{(k)}}.$$

Hence

$$\max_{j\in[n]} g_j^{(k)} \ge \frac{\mathcal{E}_k^2}{\|w\|_1^2},$$

and consequently

$$\mathcal{E}_{k+1} \le \mathcal{E}_k - \frac{\mathcal{E}_k^2}{\|w\|_1^2}.$$

Let $C = \|w\|_1^2$. Since the columns are normalized,

$$\mathcal{E}_0 = \|Aw\|_2^2 \le \left(\sum_{j=1}^{n} |w_j| \|A_j\|_2 \right)^2 = C.$$

Solving the recurrence gives

$$\mathcal{E}_k \le \frac{\mathcal{E}_0}{1 + k\mathcal{E}_0/C}.$$

Using $\mathcal{E}_0 \le C$, this also implies

$$\mathcal{E}_k \le \frac{C}{k+1} = \frac{\|w\|_1^2}{k+1}.$$

□

Proof of Theorem 2. Fix an iteration k, and condition on the current residual $R^{(k)}$. If pivot j is selected, then

$$R^{(k+1)} = R^{(k)} - \frac{R_{:,j}^{(k)} R_{j,:}^{(k)}}{R_{j,j}^{(k)}},$$

and therefore

$$\mathcal{E}_{k+1} = \mathcal{E}_k - g_j^{(k)}.$$

If all scores vanish, then the lower bound below implies $\mathcal{E}_k = 0$, so the result is immediate. Otherwise, taking conditional expectation with respect to the randomized pivot rule gives

$$\mathbb{E}\left[\mathcal{E}_{k+1} \mid R^{(k)}\right] = \mathcal{E}_k - \frac{\sum_{j=1}^{n} (g_j^{(k)})^2}{\sum_{\ell=1}^{n} g_\ell^{(k)}}.$$

By Cauchy-Schwarz,

$$\sum_{j=1}^{n} (g_j^{(k)})^2 \ge \frac{1}{n} \left(\sum_{j=1}^{n} g_j^{(k)} \right)^2,$$

so

$$\mathbb{E}\left[\mathcal{E}_{k+1} \mid R^{(k)}\right] \le \mathcal{E}_k - \frac{1}{n} \sum_{j=1}^{n} g_j^{(k)}.$$

Next we relate the sum of scores to $\mathcal{E}_k$. Since

$$\mathcal{E}_k = w^\top R^{(k)} w = \sum_{j=1}^{n} w_j R_{j,:}^{(k)} w,$$

and $(R_{j,:}^{(k)} w)^2 = g_j^{(k)} R_{j,j}^{(k)}$, we get

$$\mathcal{E}_k \le \sum_{j=1}^{n} |w_j| \sqrt{g_j^{(k)} R_{j,j}^{(k)}}.$$

Applying Cauchy-Schwarz and using $R_{j,j}^{(k)} \leq R_{j,j}^{(0)} = 1$, we obtain

$$\mathcal{E}_k^2 \leq \left(\sum_{j=1}^{n} g_j^{(k)}\right)\left(\sum_{j=1}^{n} R_{j,j}^{(k)} w_j^2\right) \leq \left(\sum_{j=1}^{n} g_j^{(k)}\right)\|w\|_2^2.$$

Thus

$$\sum_{j=1}^{n} g_j^{(k)} \geq \frac{\mathcal{E}_k^2}{\|w\|_2^2}.$$

Substituting this into the conditional expectation bound yields

$$\mathbb{E}\left[\mathcal{E}_{k+1} \mid R^{(k)}\right] \leq \mathcal{E}_k - \frac{\mathcal{E}_k^2}{n\|w\|_2^2}.$$

Taking total expectation and applying Jensen's inequality gives

$$\mathbb{E}[\mathcal{E}_{k+1}] \leq \mathbb{E}[\mathcal{E}_k] - \frac{\mathbb{E}[\mathcal{E}_k]^2}{n\|w\|_2^2}.$$

Let $e_k = \mathbb{E}[\mathcal{E}_k]$ and $C = n\|w\|_2^2$. The recurrence becomes

$$e_{k+1} \leq e_k - \frac{e_k^2}{C}.$$

Solving it gives

$$e_k \leq \frac{e_0}{1 + ke_0/C}.$$

Finally, since the columns are normalized,

$$e_0 = \mathcal{E}_0 = \|Aw\|_2^2 \leq \|w\|_1^2 \leq n\|w\|_2^2 = C.$$

Therefore,

$$\mathbb{E}[\mathcal{E}_k] = e_k \leq \frac{\mathcal{E}_0}{1 + k\mathcal{E}_0/(n\|w\|_2^2)} \leq \frac{n\|w\|_2^2}{k+1}.$$

□